

Education

University of Trento

Master's Degree in Artificial Intelligence Systems

Trento, Italy  
September 2021 - Graduated March 2025

- **Thesis:** Exploring the Use of LLMs for Agent Planning: Strengths and Weaknesses. [ [report](#) | [code](#) ]
- **Relevant Courses:** Machine Learning, Deep Learning, Natural Language Understanding, Automated Planning, Law & Ethics in AI, Fundamentals of AI (Reasoning and Planning), Signal Processing, AI for Finance, High Performance Computing.

University of Trento

Bachelor's Degree in Computer Science

Trento, Italy  
September 2017 - Graduated June 2021

- **Thesis:** Healthy Plus - Redesign and evolution of an Android application for monitoring healthy lifestyles.
- **Core Topics:** Algorithms & Data Structures, Software Engineering, Database Systems, Operating Systems, Computer Architectures, Distributed Networks.

Work Experience

OpenCity Labs - Startup

AI & Machine Learning ENGINEER - R&D

Remote  
August 2025 - Current

- Single-handedly engineered an NLP pipeline for RAG ingestion with PII anonymization via a fine-tuned Transformer-based NER model and an in-memory reversible dictionary, processing 50k+ daily inputs.
- Built a spaCy detection + LLM fallback guard to fix a cross-lingual model failure in a RAG pipeline: despite explicit English instructions, Italian-only retrieved context caused the model to default to Italian in 35-40% of non-Italian queries, eliminating the error. Released to 10+ clients.
- Developed an open-source agentic AI system. Designed hybrid RAG architecture (vector + full-text reciprocal rank fusion) to solve domain-specific retrieval failures: pure embedding search missed legal acronyms and regulatory references, while keyword search failed on semantic ones. Built MCP-based tool-calling infrastructure with human-in-the-loop approval, reducing targeted workflows from days to hours.
- Dockerized all services with GitLab CI pipelines automating testing and Docker image builds for consistent, reliable deployments.

Python, Transformers, NLP, LLM, RAG, FastAPI, Docker, SurrealDB, AWS EC2, Prometheus, Grafana

Eurecat Technology Center

INTERN / AI ENGINEER

Barcelona, Spain  
April 2024 - June 2024

- Implemented a token-level uncertainty estimation framework for planning tasks by analyzing log-probabilities, benchmarking performance across proprietary (GPT-4o) and locally quantized models (Llama 3.1 via llama.cpp).
- Developed a RAG system utilizing *gte-large* embeddings to retrieve context-aware user preferences, significantly reducing model hallucination in recurrent decision-making scenarios.
- Prototyped a multimodal agent integrating LLaVA for visual reasoning tasks and validated the uncertainty reconstruction approach across multiple simulated environments.

Python, PyTorch, Hugging Face, OpenAI API, llama.cpp, Ollama

Projects

COVID-19 Lung Ultrasound Images classification - Medical Imaging Diagnostic

[ [slides](#) | [report](#) | [code](#) ]

- Designed a Multi-stage deep learning model to classify Lung Ultrasound images based on a multiclass severity scoring (0-3).
- Developed the 3 components of the model: a multi-class frame classifier, an uncertainty detection model and a similarity module.
- Dataset composed of anonymized images from 14 real patients for a total of 47k frames used to train and fine-tune various Computer Vision models.

Python, PyTorch, scikit-learn, Pandas, NumPy, Azure, Computer Vision

Joint Intent Detection and Slot Filling - Natural Language Understanding

[ [report](#) | [code](#) ]

- Implemented 4 Deep Learning models to simultaneously learn Intent Detection and Slot Filling tasks.
- Consistently improved baseline results by at least 2% on the ATIS dataset for both tasks, and achieved a 13% improvement on the SNIPS dataset.
- Fine-tuned BERT and ERNIE; built from scratch a model based on Bidirectional LSTM and an Encoder-Decoder architecture.

Python, PyTorch, NLP (BERT/ERNIE/LSTM)

Domain Adaptation / Transfer Learning - Deep Learning - Team Project

[ [code](#) ]

- Built and evaluated an Unsupervised Domain Adaptation Computer Vision model to bridge the synthetic-to-real gap, training on rendered images to accurately classify real-world pictures.
- Solution based on ResNet34 plus a custom adaptation layer with loss based on direct 3rd-order and grouped 4th-order statistics.
- Approach tested on the Adaptope dataset, reaching a +11.53% improvement with respect to the baseline non-adapted model.

Python, PyTorch, Google Colab

Skills

|                  |                                                                                             |
|------------------|---------------------------------------------------------------------------------------------|
| Languages        | Python, JavaScript, C/C++, SQL, Bash, <del>TeX</del>                                        |
| ML & Data        | PyTorch, Hugging Face, TensorFlow, Transformers, spaCy, scikit-learn, Pandas, NumPy, OpenCV |
| DevOps & Tools   | Git, Docker, GitLab CI, SurrealDB, Grafana, Prometheus, MPI (High Performance Computing)    |
| Spoken Languages | English - Professional, Italian - Native                                                    |